



Using the Synopsys TetraMAX Model

October 2003

Release 1.0

Copyright © 1997-2003 Artisan Components, Inc. All rights reserved.
Printed in the United States of America.

Artisan Components is a registered trademark and Artisan, Process-Perfect, Universal Test Interface, SAGE, SAGE-HS, SAGE-X, SAGE-Modeler, Flex-Repair, ArtNuvo and ElectroArt are trademarks of Artisan Components, Inc. Artisan acknowledges the trademarks of other organizations for their respective products or services mentioned in this document.

Artisan reserves the right to make changes to any products and services described herein, at any time without notice in order to make improvements in design, performance, or presentation and to provide the best possible products and services. Customers should obtain the latest specifications before referencing any information, product, or service described herein, except as expressly agreed in writing by an officer of Artisan.

Artisan does not assume any responsibility or liability arising out of the application or use of any products or services described herein, except as expressly agreed to in writing by an officer of Artisan; nor does the purchase, lease, or use of a product or service from Artisan convey a license under any patent rights, copy rights, trademark rights, or any other intellectual property rights of Artisan or of third parties.

Artisan Components, Inc., 141 Caspian Court, Sunnyvale, CA 94089, USA

Unpublished – rights reserved under the copyright laws of the United States.

Revision History

Part Number	Release Number	Release Date	Comments
ev-synopsys-tetramax-1.0	1.0	October 2003	

Using the Synopsys TetraMAX Model

After generating the Synopsys TetraMAX model file, *<name>.tv*, check the models using the following steps. You can choose the instance name, *<name>*.

Generate the TetraMAX model, *<name>.tv*, with top module *<name>*.

Generate the Verilog model, *<name>.v*, with top module *<name>*.

Generate a TetraMAX script file, *tmax.scr*, with the following contents:

```
set netlist
read netlist <name>.tv
report violation -all
run build_model <name>
add clocks 0 CLK
run drc
set atpg -capture_cycles 6
remove faults -all
add fault -all
run atpg fast_sequential_only
report_faults - all
write patterns <name>.tb.v -format verilog_single -Replace
quit
```

In adding clocks to the script above (i.e., `add clocks 0 CLK`), list all possible clocks. For example, if you have dual-port SRAM or two-port register file memories, use `CLKA` and `CLKB` instead of `CLK`.

This script contains commands that run Automatic Test Pattern Generation (ATPG) on one instance of memory and generate test vectors in Verilog format. Verilog simulations can be run using the test vector file and the Verilog model of memories using any supported Verilog simulator.

The following example demonstrates a TetraMAX simulation using VCS.

To run TetraMAX in batch mode, enter:

```
tmax -shell tmax.scr > & tmax.log
```

To run a Verilog simulation using VCS, enter:

```
vcs <name>_tb.v <name>.v > vcs.log
```

The log generated by TetraMAX, tmax.log, should not include any error messages. However, there may be some warning messages. These can be ignored since they have been verified with the tool vendor. You must test the vcs.log file to verify that the VCS simulation passed without any mismatch.